Table 1. Sensitivity analysis of the hyperparameter α. Sensitivity is measured via the Coefficient of Variation ($CV = \sigma/\mu$) to ensure a fair comparison across datasets with varying base performance.

Dataset	$\alpha = 0.1$	$\alpha = 0.5$	$\alpha = 0.9$	$\alpha = 1.5$	$\alpha = 2.0$	**Sensitivity (CV)**
MATH	74.29	74.26	74.40	74.55	74.38	0.14%
GSM8K	87.58	87.71	87.64	87.76	87.68	0.07%
Olympiad	35.25	35.12	35.16	35.04	35.19	0.20%
OMNI-MATH	16.52	16.59	16.54	16.59	16.55	0.17%
Average						**0.14%**

Table 2. Sensitivity analysis of the hyperparameter β. Sensitivity is measured via the Coefficient of Variation ($CV = \sigma/\mu$) to ensure a fair comparison across datasets with varying base performance.

Dataset	$\beta = 0.1$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$	$\beta = 2.0$	**Sensitivity (CV)**
MATH	74.36	74.40	74.41	74.35	74.43	0.04%
GSM8K	87.67	87.64	87.60	87.64	87.59	0.03%
Olympiad	35.24	35.16	35.14	35.17	35.20	0.10%
OMNI-MATH	16.49	16.54	16.52	16.58	16.46	0.25%
Average						**0.11%**

Table 3. Sensitivity analysis of the hyperparameter γ in the proposed VFPO. Sensitivity is measured via the Coefficient of Variation ($CV = \sigma/\mu$) to ensure a fair comparison across datasets with varying base performance.

Dataset	$\gamma = 0.005$	$\gamma = 0.02$	$\gamma = 0.1$	$\gamma = 0.15$	$\gamma = 0.5$	**Sensitivity (CV)**
MATH	74.33	74.40	74.42	74.35	74.38	0.04%
GSM8K	87.57	87.64	87.52	87.62	87.66	0.06%
Olympiad	35.13	35.16	35.24	35.18	35.15	0.11%
OMNI-MATH	16.51	16.54	16.48	16.49	16.56	0.18%
Average						**0.10%**

Table 4. Sensitivity analysis of the hyperparameter λ^- in the proposed VFPO. Sensitivity is measured via the Coefficient of Variation ($CV = \sigma/\mu$) to ensure a fair comparison across datasets with varying base performance.

Dataset	$\lambda^- = 0.1$	$\lambda^- = 0.5$	$\lambda^- = 0.9$	$\lambda^- = 1.5$	$\lambda^- = 2.0$	**Sensitivity (CV)**
MATH	74.37	74.41	74.40	74.35	74.46	0.05%
GSM8K	87.62	87.55	87.64	87.68	87.61	0.05%
Olympiad	35.14	35.19	35.16	35.08	35.14	0.10%
OMNI-MATH	16.47	16.55	16.54	16.58	16.55	0.22%
Average						**0.11%**

Table 5. Main performance comparison across Qwen2.5-1.5B-Math and 3B models. Results for AIME 2024 and AIME 2025 are pass@32. The **Average** column represents the mean accuracy across all seven tested datasets. **Bold** indicates the best performance.

Model	**Method**	**MATH**	**GSM8K**	**Olympiad**	**OMNI-MATH**	**AMC**	**AIME24**	**AIME25**	**Average**
Qwen2.5-1.5B-Math	DisCO(L-ratio)	70.64	85.95	32.89	15.44	45.00	35.33	32.67	45.42
	DisCO(log-L)	70.81	86.12	32.94	15.72	42.50	35.67	32.00	45.11
	CLIP_Cov+QAE	70.58	86.25	33.70	15.85	45.00	35.67	32.67	45.67
	W-REINFORCE	70.32	85.67	33.86	15.69	45.00	35.33	32.00	45.41
	SIREN	71.50	86.55	34.75	16.21	50.00	36.67	33.30	47.00
	VFPO (Ours)	**74.40**	**87.64**	**35.16**	**16.54**	**55.00**	**46.67**	**36.67**	**50.30**
Qwen2.5-3B	DisCO(L-ratio)	63.71	84.26	28.54	15.53	35.00	26.00	23.00	39.43
	DisCO(log-L)	64.11	85.40	28.96	16.55	37.50	26.33	23.67	40.36
	CLIP_Cov+QAE	64.24	85.34	28.80	16.46	37.50	26.33	23.00	40.24
	W-REINFORCE	64.15	85.27	28.87	16.33	35.00	25.67	22.67	39.71
	SIREN	64.32	85.74	29.12	16.84	40.00	26.67	23.30	40.86
	VFPO (Ours)	**66.34**	**85.90**	**29.97**	**17.11**	**47.50**	**30.00**	**33.30**	**44.30**

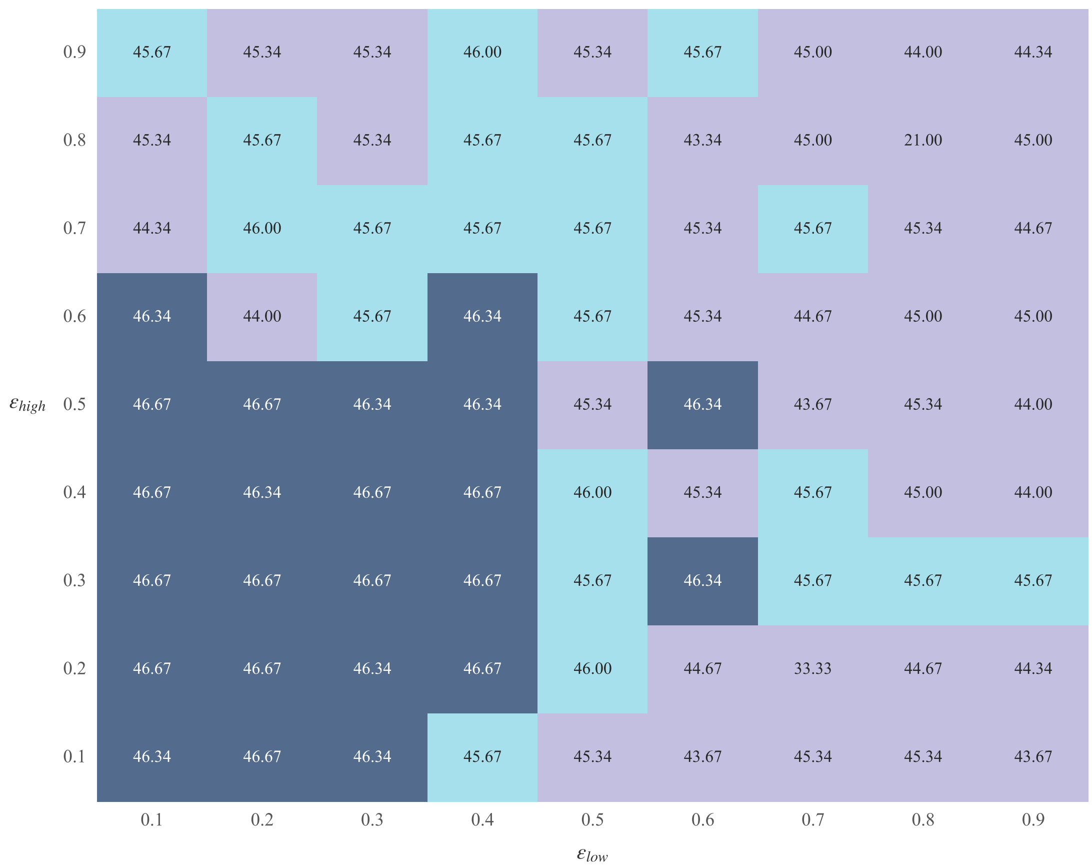

Figure 1. sensitivity analysis of ϵ_{low} and ϵ_{high}. The heatmap reveals a **broad and flat optimal plateau** (notably in the lower-left region), where the model maintains consistently high performance with minimal variance.

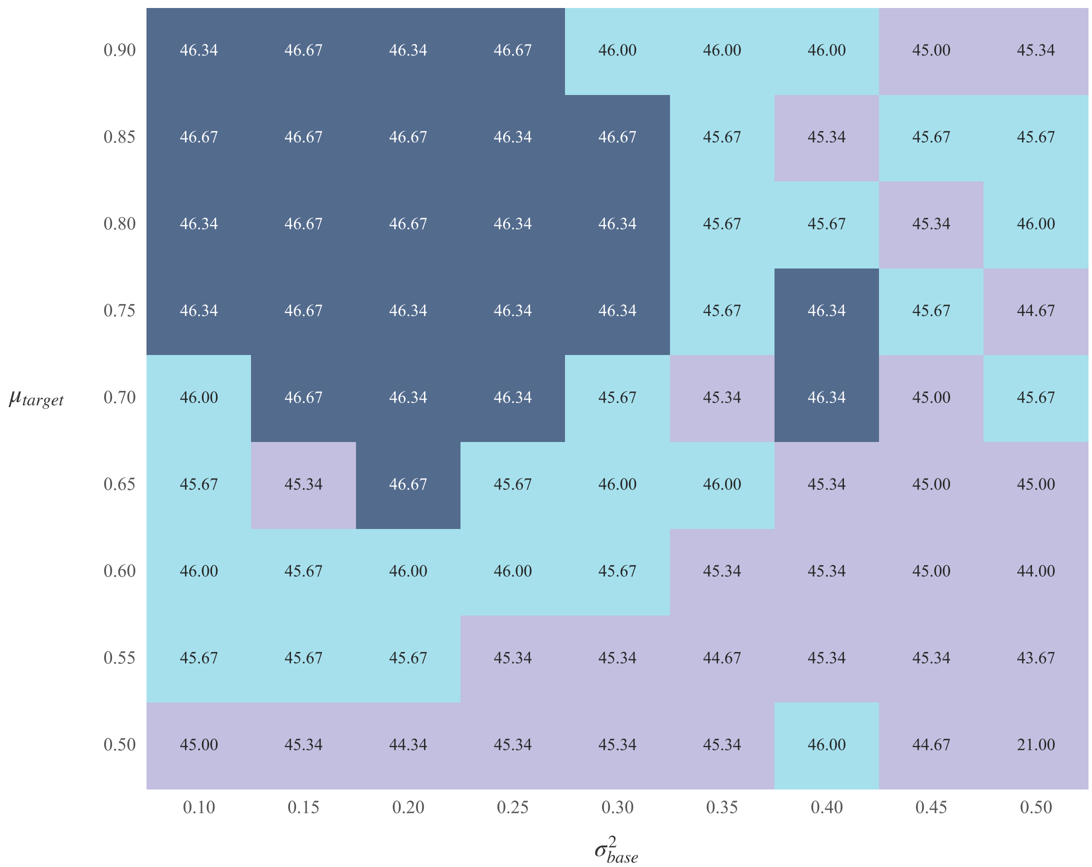

Figure 2. **Sensitivity analysis** of σ^2_{base} and μ_{target}. The heatmap illustrates a **substantial and stable peak performance region**, particularly as both parameters increase (top-left quadrant).

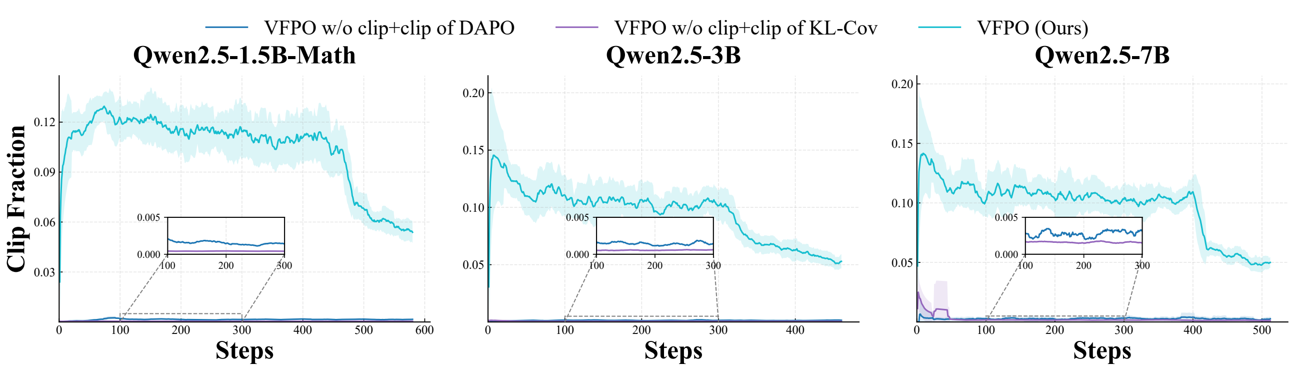

Figure 3. **Comparison of Clipping Mechanisms** To isolate the effects of different mechanisms, we build upon an **ablated VFPO baseline** (without difference-based clipping) and separately incorporate the clipping components of **DAPO**, **KL-Cov**, and our **full VFPO**.

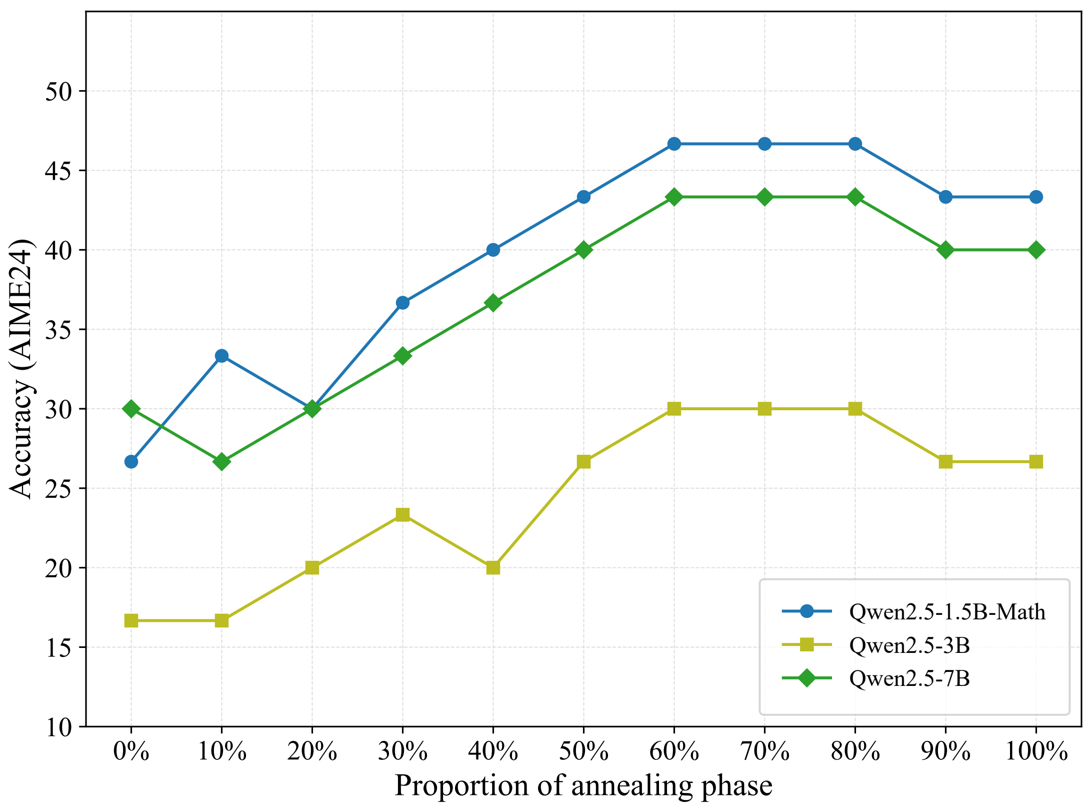

Figure 4. Performance sensitivity to the **annealing start point** (as a percentage of total training).